

JavaScript

1. DOM

- a. querySelector
- b. textContent
- c. addEventListener
- d. Order of Parsing

2. event Propagation

- a. event Bubbling
- b. event Capturing/ Trickling
- c. how to add both on program

3. event.stopPropagation();

4. inst

- a. e.target
 - i. id
 - ii. tagName
 - iii. pros and cons

5. Architecture

- a. Execution context
 - i. variable environment (memory)
 - ii. Thread of execution (code)
 - iii. - global & local execution context
 - iv. - phases
 - 1. Memory allocation
 - 2. Code execution
- b. Synchronous single threaded app
- c. Call stack
- d. Proxy
 - i. Proxy traps
 - ii. Reflect
 - iii. proxy vs reflect
- e. **Event loop**
 - i. Callback queue/ task queue
 - ii. Microtask queue
 - 1. mutation observer
 - iii. Starvation

- iv. Memory Heap
- f. Just In Time Compilation
- g. Interpreter vs Compiler
- h. Abstract Syntax Tree
- i. Concurrency model

6. Theory

7. Data types

- a. wrapper objects
- b. 0 vs new Number(0)

c. Numbers

- i. 1_000_000
- ii. 1e9, 1e-6
- iii. Hex, binary and octal numbers
- iv. toString(base)
- v. Math.trunc

8. Operators

9. enum

- a. how to get enum in javascript

10. Function

- a. Function Statement
- b. Function Expression
- c. Function Declaration
- d. Anonymous function
- e. Named Function Expression
- f. Functional Programming
- g. **Higher order function**
- h. First class function
- i. **Decorator function**
 - i. use
 - ii. - count no of function call
 - iii. - valid data of params
- j. **Pure function**
 - i. pros and cons
 - ii. rules
 - iii. pure vs impure
- k. IIFE
 - i. pros

11. Advantages and disadvantages of JS

12. Set Map Flat

- a. set

- i. add, delete, has, clear, keys, values, entries
 - ii. <setName>.size
- b. map
 - i. get, set, has, delete, clear, keys, values, entries, forEach
 - ii. iterating
- c. object vs map
- d. weekSet()
 - i. features
- e. weekMap()
 - i. features
 - ii. key is private
- f. Week set and map summary
- g. falt()
- h. flatMap()
- i. reduceRight()
- j. copyWithin()

13. Operators

- a. Nullish coalescing operator
- b. Optional chaining
- c. || vs ??
- d. Ternary operator
- e. Type Operators
- f. **Unary operators**
 - i. delete
 - ii. typeof
 - iii. !, ++, -, +
- g. **Bitwise Operators**
 - i. bitwise OR
 - ii. bitwise AND
 - iii. uses

14. Scope

- a. Global scope
- b. Module scope
- c. Function scope
- d. Lexical scope
- e. Block scope

15. Shadowing & Illegal shadowing

16. Prototype

17. Types of error

- a. syntax, logic

18. Closure

- a. Disadvantage

- b. Uses
- c. lexical scope vs closure
- d. IIFE

19. Garbage collection

- a. How does it work?
- b. mark-and-sweep
- c. reachability

d. Optimizations

- i. - Generational collection
- ii. - Incremental collection
- iv. - Idle-time collection

20. Hoisting

- a. TDZ let, const vs var
- b. Function vs arrow function

21. Call Apply Bind

- a. function borrowing
- b. call vs apply vs bind
- c. polyfills

22. transpiler

- a. Babel.
- b. webpack

23. polyfills vs transpiler

24. This Keyword

25. String Methods

- a. Length, toUpperCase, LowerCase, Trim, Pad, charAt, Split, Concat, substring, indexOf, lastIndexOf, localeCompare

26. Array Methods

- a. Map, Filter, Reduce, Find, Sort, Foreach, Push, Pop, Shift, Unshift, Slice, Splice, concat, indexOf, lastIndexOf, forEach, split, join, reduceRight, iArray, fill, copy, flat
- b. spare array, jagged array, hols in array
- c. copy within
- d. typed arrays

27. Object Methods

- a. object constructor, literal

- b. deleting field
 - c. Computed properties
 - d. __proto__
 - e. in
 - f. Object.assign
 - g. structuredClone
 - h. _cloneDeep(obj)
 - i. methods
 - j. this keyword
 - k. Symbol type
28. Symbol
- a. properties
 - b. use
 - c. ongo
 - d. global symbol registry
 - e. for, keyFor, iterator, toPrimitive
29. **Loop**
- a. for
 - b. do while vs while
 - c. labelled statements
 - d. - break
 - e. - continue
 - f. for...in
 - g. for...of
30. **Callback**
- a. callback hell
 - b. inversion of control
31. **Promises**
- a. Promise states
 - b. Promise chaining
 - c. Promise.all
 - d. Promise.allSettled
 - e. Promise.any
 - f. Promise.race
 - g. Promise.resolve
 - h. Thenable
 - i. Finally
 - j. Catch
 - k. immutable
 - l. promisify
 - m. pros and cons
32. **Async await**
- a. async always return a promise
 - b. error handling in async await
33. **Debouncing & Throttling**
- a. both are used for optimising performance of a web app
 - b. by limiting the rate of API calls
34. Spread and Rest Operator
35. DOM, BOM
36. Window Object
37. **ES6 and its features**
- a. Let, Var, Const
 - b. Ternary operator
 - c. Arrow function
 - d. Template literals
 - e. Default Parameters
 - f. Classes
 - g. Modules
 - h. Iterators
 - i. Object & Array Destructuring
38. **Primitive and non-primitive**
- a. Pass by value and pass by reference
39. Message queue
40. Life
41. Generator
42. **Prototype**
- a. Prototype chain
 - b. Prototypal Inheritance
 - c. uses?
 - d. Circular reference
 - e. Object.key
43. **Recursion**
- a. recursive call to function
 - b. condition to exit
 - c. pros and cons
 - d. display the fibonacci sequence
 - e. use
44. JavaScript is dynamically types
45. **Currying**
- a. function inside function
46. **Type Casting**
- a. Implicite (Coercion)
 - b. Explicit (Conversion)
47. Microtask queue

48. **Shallow copy vs Deep copy**

- a. primitive vs structural
- b. how to make these copies
- c. pros and cons
- d. Mutable vs Immutable
- e. Object.freeze()

49. TCP/IP

50. DNS

51. **IIFE**

- a. pros and cons

52. **Composition vs Inheritance**

53. Function recursion

54. [Symbol.iterator]

55. Truthy and falsy value

56. Strict mode in JS

57. this substitution

58. **VS**

- a. label vs func
- b. == and ===
- c. Let, const, var
- d. Synchronous vs asynchronous
- e. While vs do while
- f. Foreach Vs Map
- g. Parameters, Arguments
- h. for in, for of
- i. Undefined, Null
- j. Keywords & Identifiers
- k. Type casting vs Type coercion
- l. textContent vs innerText
- m. identifiers vs variables
- n. defer vs async

59. **Good to Know**

60. interpreted and compiled doe

61. Server-side vs client-side code

62. with in js